Date: Day:

Deep RNNS :- (Stacked)

D.R is stacking 1 RNN cell on top of another & then unfolding all of them in time

Why & when to use:-

a) Hierarchial representation.

b) Customization for adv. tasks.

D.R.N.N used for adv. level encoder, decoder.

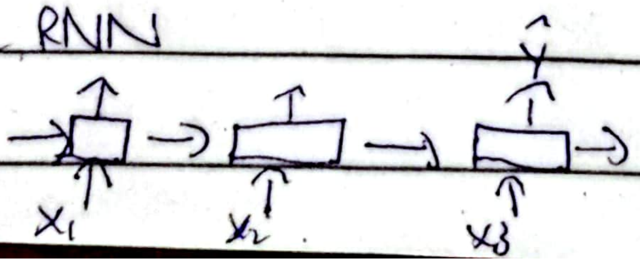
review	sentiment
cat mat rat	1
rat rat mat	0
mat cat cat	1

When to use:-

- Complex tasks
  - speech recogn
  - Machine translation
- Large datasets
- Computational
  - Not satisfied simpler model

Bidirectional RNN:-

Simple RNN



Date: Day:

But let's say NER (Name entity recognition)

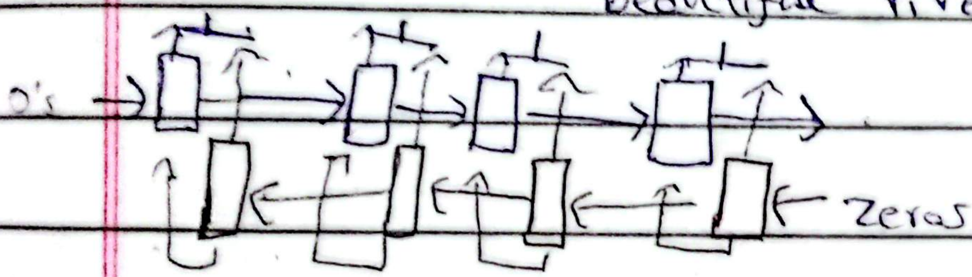
- I love $\boxed{\text{amazon}}$ it's a great website
- I love $\boxed{\text{amazon}}$ it's a beautiful river.

So here in cases like this we use bidirectional RNN's

Bidirectional RNN architecture:-

Amazon the best website.

" " beautiful river,



Amazon the best website.

$$\vec{h}_t = \tanh(\vec{W} h_{t-1} + \vec{U} x_t + \vec{b})$$

$$\overleftarrow{h}_t = \tanh(\overleftarrow{W} h_{t+1} + \overleftarrow{U} x_t + \overleftarrow{b})$$

$$y_t = \sigma(V(\vec{h}_t, \overleftarrow{h}_t) + b)$$

All things exactly same just add Bidirectional(SimpleRNN(5)), in keras.

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- In case of lstm known as bilstm.
- " " of GRU " " biGRU.

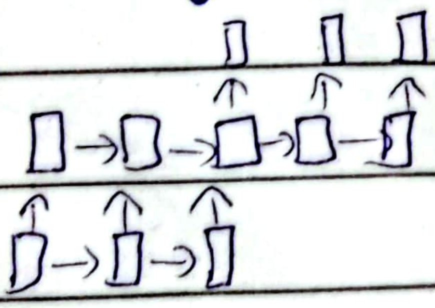
Applications & drawbacks:-

NER
Pos tagging
machine translation.
Sentiment analysis.
Timeseries forecasting.

→ Drawback is complexity (weights & biases b/m double).

→ Real time speech recog. → latency issues.

History of LLM's:-



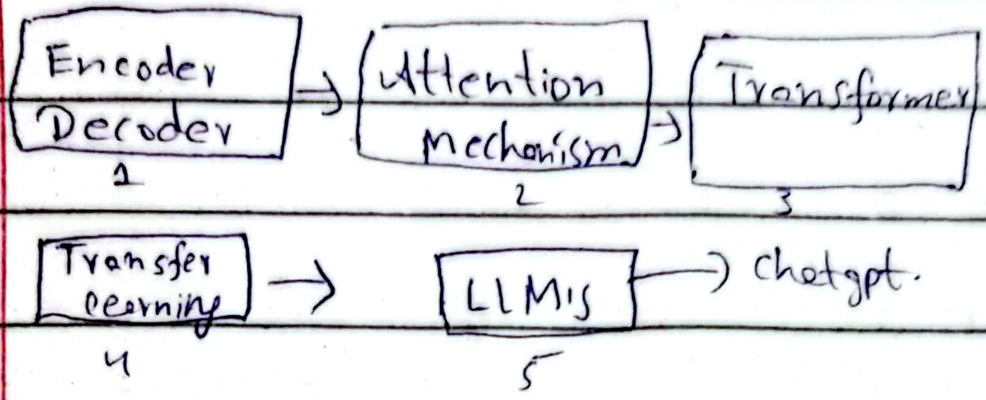
Seq to seq models.

→ many to many asynchronized.

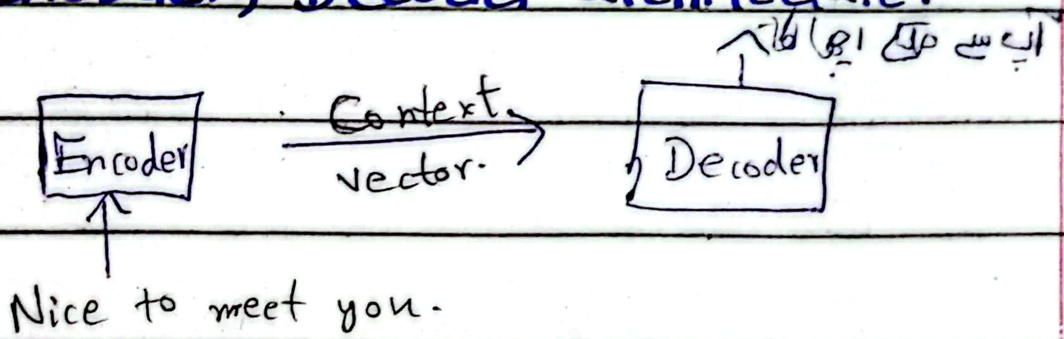
1) Text summarization.

2) Question, answering.

CS CamScanner



Encoder, Decoder architecture:-



Attention mechanism:-

Problem in encod, decoder is when encoder process the sentence genciate a summery, but sometimes when sentence b/m too large ∂ it fails.

→ In decoder we pass on that which time step is useful of encoder.